

18/04/2024

Phase-I
CODE-A

Aakash

Medical | IIT-JEE | Foundations

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FINAL TEST SERIES for NEET-2024

MM : 720

Test-9

Time : 3 Hrs. 20 Mins.

Complete Syllabus of Class XI & XII

Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION-A

- A body is executing simple harmonic motion with frequency ' n ', the frequency of its potential energy is
 - $4n$
 - n
 - $2n$
 - $3n$
- Polar molecules are the molecules
 - Having a permanent electric dipole moment
 - Having zero dipole moment
 - Acquire a dipole moment only in the presence of electric field due to displacement of charges
 - Acquire a dipole moment only when magnetic field is absent
- Column-I** gives certain physical terms associated with flow of current through a metallic conductor.

Column-II gives some mathematical relations involving electrical quantities. Match **Column-I** and **Column-II** with appropriate relations.

	Column-I		Column-II
(A)	Drift Velocity (V_d)	(P)	$\frac{m}{ne^2\rho}$
(B)	Electrical Resistivity (ρ)	(Q)	nev_d
(C)	Relaxation Period (τ)	(R)	$\frac{eE}{m}\tau$
(D)	Current Density (J)	(S)	$\frac{E}{J}$

- (A) - (R), (B) - (Q), (C) - (S), (D) - (P)
- (A) - (R), (B) - (S), (C) - (P), (D) - (Q)
- (A) - (R), (B) - (S), (C) - (Q), (D) - (P)
- (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

4. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is

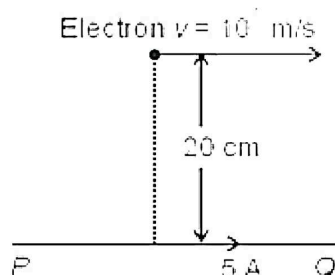
(1) $\frac{R_1^2}{R_2^2}$ (2) $\frac{R_1}{R_2}$
 (3) $\frac{R_2}{R_1}$ (4) $\sqrt{\frac{R_1}{R_2}}$

5. A parallel plate capacitor has a uniform electric field $\vec{E}$ in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is

(ϵ_0 = permittivity of free space)

(1) $\frac{E^2 Ad}{\epsilon_0}$ (2) $\frac{1}{2} \epsilon_0 E^2$
 (3) $\epsilon_0 E A d$ (4) $\frac{1}{2} \epsilon_0 E^2 A d$

6. An infinitely long straight wire carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



(1) 8×10^{-20} N (2) 4×10^{-20} N
 (3) $8\pi \times 10^{-20}$ N (4) $4\pi \times 10^{-20}$ N

7. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of

(1) $[M^2] [L^{-2}] [T^{-1}]$ (2) $[M^2] [L^{-1}] [T^0]$
 (3) $[M] [L^{-1}] [T^{-1}]$ (4) $[M] [L^0] [T^0]$

8. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since

- (1) A large aperture contributes to the quality and visibility of the images.
 (2) A large aperture of the objective ensures better light gathering power.
 (3) A large aperture provides a better resolution.
 (4) All of the above

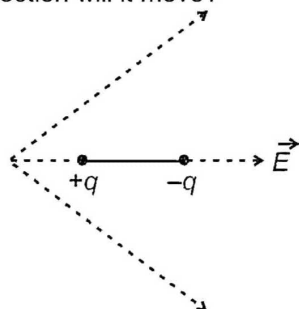
9. Match **Column - I** and **Column - II** and choose the correct match from the given choices.

	Column - I		Column - II
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3} nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3 RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2} RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2} k_B T$

- (1) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)
 (2) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
 (3) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)
 (4) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)

Space for Rough Work

10. Consider the following **statements (A)** and **(B)** and identify the **correct** answer.
(A) A zener diode is connected in reverse bias, when used as a voltage regulator.
(B) The potential barrier of p - n junction lies between 0.1 V to 0.3 V.
- (1) **(A)** is incorrect but **(B)** is correct
 (2) **(A)** and **(B)** both are correct
 (3) **(A)** and **(B)** both are incorrect
 (4) **(A)** is correct and **(B)** is incorrect
11. A dipole is placed in an electric field as shown. In which direction will it move?



- (1) Towards the right as its potential energy will increase
 (2) Towards the left as its potential energy will increase
 (3) Towards the right as its potential energy will decrease
 (4) Towards the left as its potential energy will decrease
12. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be
- (1) 30 (2) 25
 (3) 15 (4) 50
13. The escape velocity from the Earth's surface is v . The escape velocity from the surface of another planet having radius four times that of Earth and same mass density is
- (1) $4v$ (2) v
 (3) $2v$ (4) $3v$

14. A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence ${}^A_ZX \rightarrow {}^{Z-1}_B \rightarrow {}^{Z-3}_C \rightarrow {}^{Z-2}_D$ where Z is the atomic number of element X . The possible decay particles in the sequence are

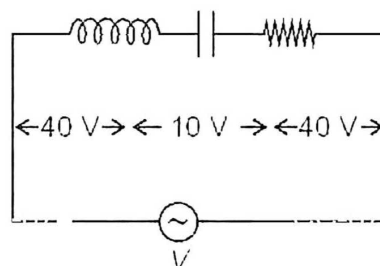
- (1) β^- , α , β^+ (2) α , β^- , β^+
 (3) α , β^+ , β^- (4) β^+ , α , β^-

15. A screw gauge gives the following readings when used to measure the diameter of a wire
 Main scale reading : 0 mm
 Circular scale reading : 52 divisions
 Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is

- (1) 0.052 cm (2) 0.52 cm
 (3) 0.026 cm (4) 0.26 cm

16. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is



- (1) 5Ω (2) $4\sqrt{2} \Omega$
 (3) $\frac{5}{\sqrt{2}} \Omega$ (4) 4Ω

Space for Rough Work

17. A particle is released from height H from the ground. At a certain height its kinetic energy is three times its potential energy. The height from the ground and the speed of the particle at that instant are respectively

(1) $\frac{H}{4}, \sqrt{\frac{3gH}{2}}$ (2) $\frac{H}{4}, \frac{3gH}{2}$
 (3) $\frac{H}{4}, \frac{\sqrt{3gH}}{2}$ (4) $\frac{H}{2}, \frac{\sqrt{3gH}}{2}$

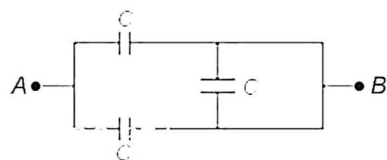
18. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is

(1) $\frac{2n}{2n-1}$ (2) $\frac{2n-1}{2n}$
 (3) $\frac{2n-1}{2n+1}$ (4) $\frac{2n+1}{2n-1}$

19. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$
 (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{2}{3}$

20. The equivalent capacitance of the combination between the terminal AB as shown in the figure is



(1) $\frac{3C}{2}$ (2) $3C$
 (3) $2C$ (4) $\frac{C}{2}$

21. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω . What will be the effective resistance if they are connected in series?

(1) 4Ω (2) 0.25Ω
 (3) 0.5Ω (4) 1Ω

22. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of parent nuclei is 7.6 MeV while that of daughter nuclei is 8.5 MeV . The total gain in the binding energy in the process is

(1) 216 MeV
 (2) 0.9 MeV
 (3) 9.4 MeV
 (4) 804 MeV

23. The electron concentration in an n -type semiconductor is the same as hole concentration in a p -type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

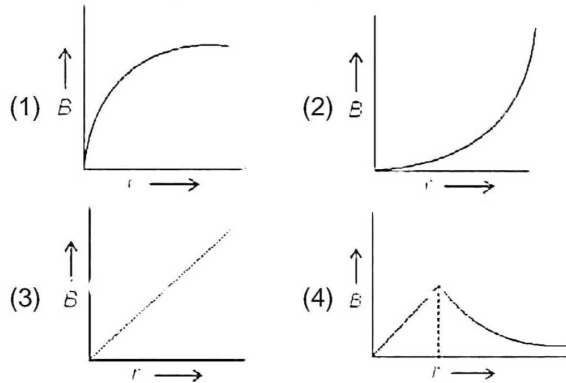
(1) No current will flow in p -type, current will only flow in n -type
 (2) Current in n -type = current in p -type
 (3) Current in p -type > current in n -type
 (4) Current in n -type > current in p -type.

24. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm , when it delivers the power of $3.3 \times 10^{-3} \text{ watt}$ will be ($h = 6.6 \times 10^{-34} \text{ J s}$)

(1) 10^{15} (2) 10^{18}
 (3) 10^{17} (4) 10^{16}

Space for Rough Work

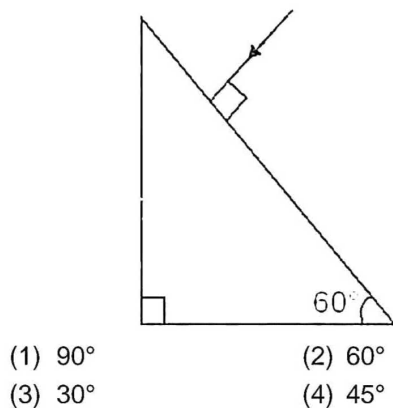
25. A thick current carrying cable of radius ' R ' carries current ' I ' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance ' r ' from the axis of the cable is represented by



26. The velocity of a small ball of mass M and density d , when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be

- (1) $2Mg$ (2) $\frac{Mg}{2}$
(3) Mg (4) $\frac{3}{2}Mg$

27. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



28. A capacitor of capacitance ' C ', is connected across an ac source of voltage V , given by

$$V = V_0 \sin \omega t$$

The displacement current between the plates of the capacitor, would then be given by

- (1) $I_d = (V_0 \omega C) \sin \omega t$ (2) $I_d = (V_0 \omega C) \cos \omega t$
(3) $I_d = \left(\frac{V_0}{\omega C} \right) \cos \omega t$ (4) $I_d = \left(\frac{V_0}{\omega C} \right) \sin \omega t$

29. If force $[F]$, acceleration $[A]$ and time $[T]$ are chosen as the fundamental physical quantities. Find the dimensions of energy.

- (1) $[F][A^{-1}][T]$ (2) $[F][A][T]$
(3) $[F][A][T^2]$ (4) $[F][A][T^{-1}]$

30. A copper wire of radius 1 mm contains 10^{22} free electrons per cubic meter. The drift velocity of free electrons when 10 A current flows through the wire will be (Given, charge on electron = 1.6×10^{-19} C)

- (1) $\frac{6.25}{\pi} \text{ ms}^{-1}$ (2) $\frac{6.25 \times 10^5}{\pi} \text{ ms}^{-1}$
(3) $\frac{6.25 \times 10^4}{\pi} \text{ ms}^{-1}$ (4) $\frac{6.25}{\pi} \times 10^3 \text{ ms}^{-1}$

31. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is

- (1) 0.628 s (2) 0.0628 s
(3) 6.28 s (4) 3.14 s

32. For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively?

- (1) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$ (2) $\hat{j} + \hat{k}, \hat{j} + \hat{k}$
(3) $-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$ (4) $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$

Space for Rough Work

33. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is

- (1) $\frac{5}{13}t$
- (2) $\frac{13}{10}t$
- (3) $\frac{13}{5}t$
- (4) $\frac{10}{13}t$

34. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. The momentum of emitted photoelectron is

- (1) $\sqrt{\frac{mhc}{\lambda}}$
- (2) $\frac{mhc}{\lambda}$
- (3) $\frac{2mhc}{\lambda}$
- (4) $\sqrt{\frac{2mhc}{\lambda}}$

35. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine?

($g = 10 \text{ m/s}^2$)

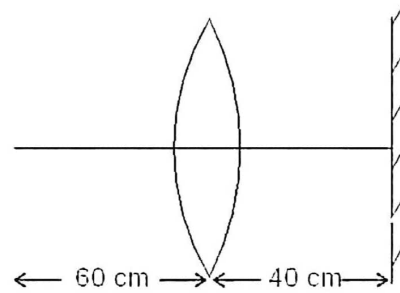
- (1) 7.0 kW
- (2) 10.2 kW
- (3) 8.1 kW
- (4) 12.3 kW

SECTION-B

36. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

- (1) 1980 V
- (2) 660 V
- (3) 1320 V
- (4) 1520 V

37. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of



- (1) 20 cm from the plane mirror, it would be a virtual image
 - (2) 20 cm from the lens, it would be a real image
 - (3) 30 cm from the lens, it would be a real image
 - (4) 30 cm from the plane mirror, it would be a virtual image
38. A particle of mass ' m ' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth.

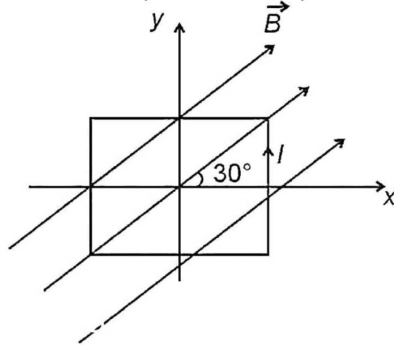
(V_e = escape velocity)

The maximum height above the surface reached by the particle is

- (1) $\frac{Rk^2}{1-k^2}$
- (2) $R\left(\frac{k}{1-k}\right)^2$
- (3) $R\left(\frac{k}{1+k}\right)^2$
- (4) $\frac{R^2k}{1+k}$

Space for Rough Work

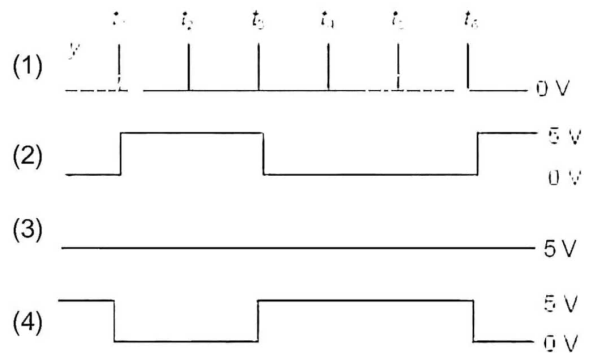
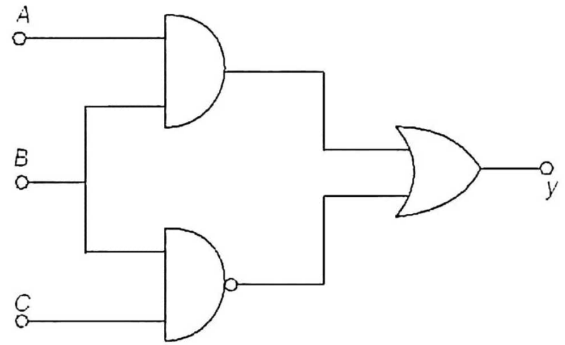
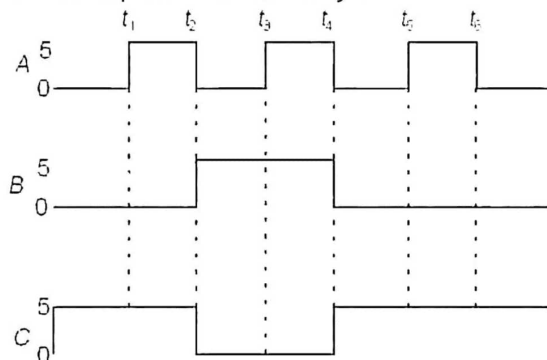
39. A current carrying square loop of side L is placed in uniform magnetic field B_0 in xy plane. The loop carries a current I in the direction shown. The magnitude of torque on the loop is



- (1) IL^2B_0 (2) $\frac{IL^2B_0}{2}$
 (3) $\frac{IL^2B_0\sqrt{3}}{2}$ (4) $2IL^2B_0$
40. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to

- (1) $\frac{R_2^2}{R_1}$ (2) $\frac{R_1}{R_2}$
 (3) $\frac{R_2}{R_1}$ (4) $\frac{R_1^2}{R_2}$

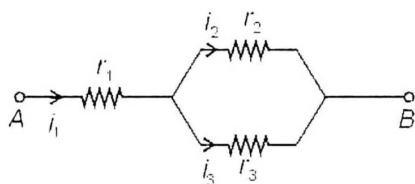
41. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y?



42. From a circular ring of mass ' M ' and radius ' R ' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is ' K ' times ' MR^2 '. Then the value of ' K ' is

- (1) $\frac{1}{8}$
 (2) $\frac{3}{4}$
 (3) $\frac{7}{8}$
 (4) $\frac{1}{4}$

43. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is

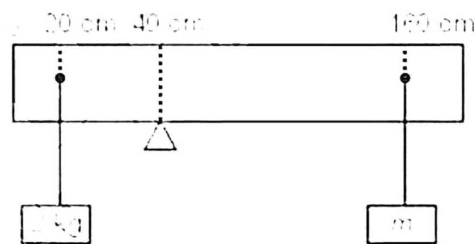


- (1) $\frac{r_2}{r_1 + r_3}$ (2) $\frac{r_1}{r_2 + r_3}$
 (3) $\frac{r_2}{r_2 + r_3}$ (4) $\frac{r_1}{r_1 + r_2}$
44. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$?
 (Take $g = 10 \text{ m/s}^2$)
 (1) $20\sqrt{2} \text{ m/s}$, 10 m/s^2
 (2) 20 m/s , 5 m/s^2
 (3) 20 m/s , 0
 (4) $20\sqrt{2} \text{ m/s}$, 0
45. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V , 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?
 (1) 4 A (2) 0.2 A
 (3) 0.4 A (4) 2 A
46. A ball of mass 0.15 kg is dropped from a height 10 m , strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly
 (1) 1.4 kg m/s (2) 0 kg m/s
 (3) 4.2 kg m/s (4) 2.1 kg m/s

47. A series LCR circuit containing 5.0 H inductor, $80 \mu\text{F}$ capacitor and 40Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be

- (1) 42 rad/s and 58 rad/s
 (2) 25 rad/s and 75 rad/s
 (3) 50 rad/s and 25 rad/s
 (4) 46 rad/s and 54 rad/s

48. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)



- (1) $\frac{1}{12} \text{ kg}$
 (2) $\frac{1}{2} \text{ kg}$
 (3) $\frac{1}{3} \text{ kg}$
 (4) $\frac{1}{6} \text{ kg}$

49. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution. If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ , is then given by :

$$(1) \theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{\frac{1}{2}} \quad (2) \theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{\frac{1}{2}}$$

$$(3) \theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{\frac{1}{2}} \quad (4) \theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{\frac{1}{2}}$$

50. A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of an equilateral triangle of side a . The magnetic moment of coil is

$$(1) 2/a^2 \quad (2) \sqrt{3} / a^2$$

$$(3) 2\sqrt{3} / a^2 \quad (4) 3/a^2$$

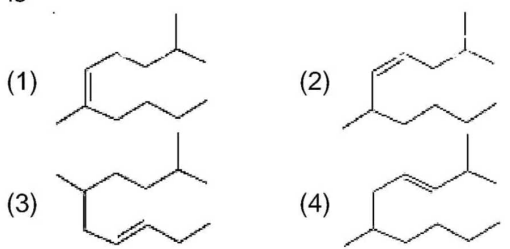
CHEMISTRY

SECTION-A

51. 1.0 g of magnesium is burnt with 0.32 g O_2 in a closed vessel. Which reactant is left in excess and how much? (At. wt. Mg = 24; O_2 = 16)
- (1) Mg, 0.52 g (2) O_2 , 0.16 g
(3) Mg, 0.68 g (4) O_2 , 0.2 g
52. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of :
- (1) Having similar chemical properties
(2) Belonging to same group
(3) Diagonal relationship
(4) Lanthanoid contraction
53. Find out the solubility of $Ni(OH)_2$ in 0.2 M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}
- (1) 4×10^{-8} M (2) 5×10^{-14} M
(3) 2×10^{-8} M (4) 4×10^{-14} M
54. Conjugate base for Bronsted acids H_2O and HCO_3^- are
- (1) OH^- and CO_3^{2-} respectively
(2) H_3O^+ and CO_3^{2-} respectively
(3) OH^- and H_2CO_3 respectively
(4) H_3O^+ and H_2CO_3 respectively
55. Correct relationship between C_p and C_v for one mole of ideal gas is
- (1) $C_v = RC_p$ (2) $C_p + C_v = R$
(3) $C_p - C_v = R$ (4) $C_p = RC_v$

56. The standard electrode potential ($E^\ominus$) values of A^{3+}/A , B^+/B , C^+/C and D^{3+}/D are -1.5 V, 0.80 V, -2.73 V and -0.73 respectively. The correct decreasing order of reducing power of the metal is
- (1) $A > B > C > D$ (2) $C > A > D > B$
(3) $D > C > A > B$ (4) $B > D > A > C$
57. Which of the following ions give smell of rotten eggs with dil. H_2SO_4 ?
- (1) CO_3^{2-} (2) SO_3^{2-}
(3) S^{2-} (4) NO_2^-
58. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on
- (1) Huckel's Rule (2) Saytzeff's Rule
(3) Hund's Rule (4) Hofmann Rule
59. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are :
- (1) sp^2 and 4 (2) sp^3 and 4
(3) sp^2 and 6 (4) sp^2 and 6
60. What is the IUPAC name of the organic compound formed in the following chemical reaction?
- Acetone $\xrightarrow[\text{(ii) } H_2O, H^+]{\text{(i) } C_2H_5MgBr, \text{ dry Ether}}$ Product
- (1) 2-Methylbutan-2-ol (2) 2-Methylpropan-2-ol
(3) Pentan-2-ol (4) Pentan-3-ol

Space for Rough Work

61. The RBC deficiency is deficiency disease of :
 (1) Vitamin B₂ (2) Vitamin B₁₂
 (3) Vitamin B₆ (4) Vitamin B₁
62. The molar conductance of NaCl, HCl and CH₃COONa at infinite dilution are 126.45, 426.16 and 91.0 S cm² mol⁻¹ respectively. The molar conductance of CH₃COOH at infinite dilution is
 (1) 540.48 S cm² mol⁻¹
 (2) 201.28 S cm² mol⁻¹
 (3) 390.71 S cm² mol⁻¹
 (4) 698.28 S cm² mol⁻¹
63. The incorrect statement among the following is :
 (1) Actinoids are highly reactive metals, especially when finely divided
 (2) Actinoid contraction is greater for element to element than lanthanoid contraction
 (3) Most of the trivalent Lanthanoid ions are colorless in the solid state
 (4) Lanthanoids are good conductors of heat and electricity
64. Given below are two statements.
Statement I : Acid strength increases in the order given as HF < HCl < HBr < HI.
Statement II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.
 In the light of the above statements, choose the correct answer from the options given below.
 (1) Statement I is false but statement II is true
 (2) Both statement I and statement II are true
 (3) Both Statement I and statement II are false
 (4) Statement I is true but statement II is false
65. The correct structure of 2,6-Dimethyl-dec-4-ene is

66. The correct sequence of bond enthalpy of 'C—X' bond is :
 (1) CH₃—Cl > CH₃—F > CH₃—Br > CH₃—I
 (2) CH₃—F < CH₃—Cl < CH₃—Br < CH₃—I
 (3) CH₃—F > CH₃—Cl > CH₃—Br > CH₃—I
 (4) CH₃—F < CH₃—Cl > CH₃—Br > CH₃—I
67. A particular station of All India Radio, New Delhi broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is: [speed of light c = 3.0 × 10⁸ ms⁻¹]
 (1) 21.92 cm (2) 219.3 m
 (3) 219.2 m (4) 2192 m
68. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is :
 (1) 6.25 (2) 8.50
 (3) 5.50 (4) 7.75
69. The following solutions were prepared by dissolving 10 g of glucose (C₆H₁₂O₆) in 250 ml of water (P₁), 10 g of urea (CH₄N₂O) in 250 ml of water (P₂) and 10 g of sucrose (C₁₂H₂₂O₁₁) in 250 ml of water (P₃). The right option for the decreasing order of osmotic pressure of these solutions is :
 (1) P₃ > P₁ > P₂ (2) P₂ > P₁ > P₃
 (3) P₁ > P₂ > P₃ (4) P₂ > P₃ > P₁
70. Given below are two statements.
Assertion (A): Boiling point of oxygen is higher than Neon.
Reason (R): Only type of interatomic interaction in noble gases is weak dispersion forces.
 In the light of above statements choose the correct answer.
 (1) Both (A) and (R) are false
 (2) (A) is true (R) is false
 (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

Space for Rough Work

71. Given below are two statements.

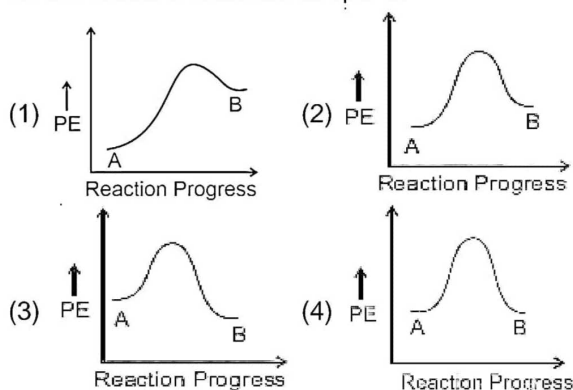
Assertion (A): More heat is required to vaporize 1 mole of water than 1 mole of acetone.

Reason (R): For an organic liquid, such as acetone, the intermolecular dipole-dipole interactions are significantly weaker.

In the light of above statements choose the **correct** answer.

- (1) Both (A) and (R) are false
- (2) (A) is true (R) is false
- (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

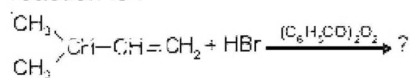
72. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



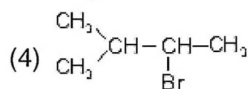
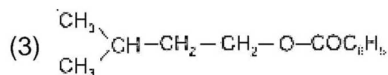
73. The compound which shows metamerism is :

- (1) $\text{C}_4\text{H}_{10}\text{O}$
- (2) C_5H_{12}
- (3) $\text{C}_3\text{H}_8\text{O}$
- (4) $\text{C}_3\text{H}_6\text{O}$

74. The major product of the following chemical reaction is :



- (1) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CBr}-\text{CH}_2-\text{CH}_3 \end{array}$
- (2) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_2-\text{Br} \end{array}$



75. Which of the following reactions is the metal displacement reaction?

- (1) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$
- (2) $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
- (3) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
- (4) $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$

76. Given below are two statements.

Assertion (A): Aromatic carboxylic acids do not undergo Friedel-Crafts reaction.

Reason (R): Carboxylic acid is deactivating and the catalyst aluminium chloride (Lewis acid) gets bonded to the carboxyl group.

In the light of above statements choose the **correct** answer.

- (1) Both (A) and (R) are false
- (2) (A) is true (R) is false
- (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

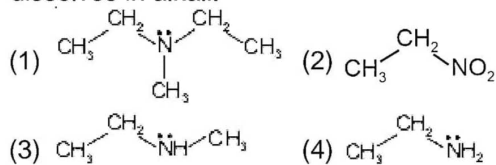
77. Ethylene diaminetetraacetato (EDTA^{4-}) ion is :

- (1) Tridentate ligand with three "N" donor atoms
- (2) Hexadentate ligand with four "O" and two "N" donor atoms
- (3) Unidentate ligand
- (4) Bidentate ligand with two "N" donor atoms

78. Strontium sulphate is mixed with conc. HCl and flame test is performed. The colour of flame is

- (1) Yellow
- (2) Violet
- (3) Green
- (4) Crimson red

79. Identify the compound that will react with Hinsberg's reagent to give a product which dissolves in alkali.



Space for Rough Work

80. Lassaigne's test is used in qualitative analysis to detect
 (1) Nitrogen (2) Boron
 (3) Oxygen (4) Potassium
81. Dihedral angle of least stable conformer of Ethane is :
 (1) 0° (2) 30°
 (3) 45° (4) 60°
82. An organic compound contains 80% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]
 (1) CH_4 (2) CH
 (3) C_2H_2 (4) CH_3
83. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them.
 (1) Noble gases have large positive values of electron gain enthalpy
 (2) Noble gases are sparingly soluble in water
 (3) Noble gases have very high melting and boiling points
 (4) Noble gases have weak dispersion forces
84. $\text{Salt} + \text{AgNO}_3 \rightarrow \text{Yellow ppt}$
 (A)
 'A' is almost insoluble in concentrated ammonium hydroxide solution. Identify the anion in the salt.
 (1) Cl^- (2) S^{2-}
 (3) I^- (4) CH_3COO^-
85. Match List-I with List-II.

	List-I (Compound)		List-II (Shape)
(a)	PCl_5	(i)	Square pyramidal
(b)	SF_6	(ii)	Trigonal planar
(c)	BrF_5	(iii)	Octahedral
(d)	BF_3	(iv)	Trigonal bipyramidal

Choose the correct answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

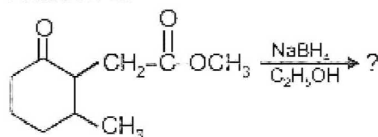
SECTION-B

86. For irreversible expansion of an ideal gas under isothermal condition, the correct option is:
 (1) $\Delta U \neq 0, \Delta S_{\text{total}} = 0$ (2) $\Delta U = 0, \Delta S_{\text{total}} = 0$
 (3) $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$ (4) $\Delta U = 0, \Delta S_{\text{total}} \neq 0$
87. Match List-I with List-II.

	List-I (Ion)		List-II (Magnetic moment)
(a)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(b)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(c)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(d)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the correct answer from the options given below.

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (3) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
88. The product formed in the following chemical reaction is:

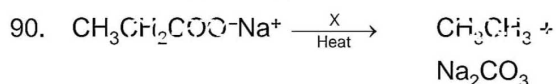


- (1)
- (2)
- (3)
- (4)

Space for Rough Work

89. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it?

- (1) $\text{CO}_2 < \text{SiO}_2$: Increasing
 $< \text{SnO}_2 < \text{PbO}_2$ oxidizing power
 (2) $\text{SbH}_3 < \text{AsH}_3$: Increasing bond
 $< \text{PH}_3 < \text{NH}_3$ angle
 (3) $\text{H}_2\text{O} < \text{H}_2\text{S}$: Increasing pK_a
 $< \text{H}_2\text{Se} < \text{H}_2\text{Te}$ values
 (4) $\text{NH}_3 < \text{PH}_3$: Increasing
 $< \text{AsH}_3 < \text{SbH}_3$ acidic character



Consider the above reaction and identify the missing reagent/chemical (X).

- (1) DIBAL-H (2) B_2H_6
 (3) Red Phosphorus (4) $\text{CaO} + \text{NaOH}$

91. Which of the following molecules is non-polar in nature?

- (1) NO_2 (2) POCl_3
 (3) CH_2O (4) SbCl_5

92. In the analysis of basic radicals, the group reagent H_2S gas is generally used in

- (1) I and II groups (2) II and III groups
 (3) III and V groups (4) II and IV groups

93. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid? Choose the **correct** option.

$$\left[\begin{array}{l} \Lambda_{\text{H}^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda_{\text{CH}_3\text{COO}^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

- (1) $2.50 \times 10^{-5} \text{ mol L}^{-1}$ (2) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
 (3) $2.50 \times 10^{-4} \text{ mol L}^{-1}$ (4) $1.75 \times 10^{-5} \text{ mol L}^{-1}$

94. When initial concentration of the reactant is doubled, its half-life period is also doubled. The order of the reaction is

- (1) 0.5 (2) Zero
 (3) First (4) Second

95. The vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is

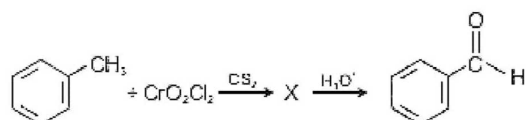
[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume ideal solution]

- (1) 350 mm of Hg (2) 160 mm of Hg
 (3) 168 mm of Hg (4) 336 mm of Hg

96. From the following pairs of ions which one is not an iso-electronic pair?

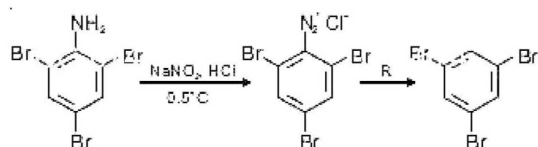
- (1) Fe^{2+} , Mn^{2+} (2) O^{2-} , F^-
 (3) Na^+ , Mg^{2+} (4) Co^{3+} , Fe^{2+}

97. The intermediate compound 'X' in the following chemical reaction is:



- (1)
- (2)
- (3)
- (4)

98. The reagent 'R' in the given sequence of chemical reaction is:



- (1) CuCN/KCN (2) H_2O
 (3) $\text{CH}_3\text{CH}_2\text{OH}$ (4) KI

Space for Rough Work

99. The slope of Arrhenius plot $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is (Choose the correct option for your answer.)

[Given $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$]

- (1) -83 kJ mol^{-1} (2) 41.5 kJ mol^{-1}
(3) 83.0 kJ mol^{-1} (4) 166 kJ mol^{-1}

100. Match List-I with List-II

	List-I (Reactions)		List-II (Name)
a.	 $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$	(i)	Hell-Volhard-Zelinsky

b.	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$	(ii)	Gatterman-Koch
c.	$\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$	(iii)	Haloform reaction
d.	$\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$	(iv)	Esterification

Choose the **correct** answer from the options given below.

- (1) a(ii), b(iii), c(iv), d(i) (2) a(iv), b(i), c(ii), d(iii)
(3) a(iii), b(ii), c(i), d(iv) (4) a(i), b(iv), c(iii), d(ii)

BOTANY

SECTION-A

101. Which of the following statements is **not** correct?
(1) Pyramid of number in a grassland ecosystem is upright
(2) Pyramid of biomass in sea is generally inverted
(3) Pyramid of biomass in sea is generally upright
(4) Pyramid of energy is always upright
102. In which one of the following, both autogamy and geitonogamy are prevented?
(1) Maize (2) Wheat
(3) Papaya (4) Castor
103. The first stable product of CO_2 fixation in *Sorghum* is
(1) Phosphoglyceric acid
(2) Pyruvic acid
(3) Oxaloacetic acid
(4) Succinic acid
104. Diadelphous stamens are found in
(1) China rose and *Citrus*
(2) China rose
(3) *Citrus*
(4) Pea

105. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as
(1) Bright blue bands
(2) Yellow bands
(3) Bright orange bands
(4) Dark red bands
106. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called
(1) Maturity (2) Elasticity
(3) Flexibility (4) Plasticity
107. Match **List-I** with **List-II**.
- | List-I | List-II |
|------------------------------|-----------------------|
| a. Parietal placentation | (i) Marigold |
| b. Axile placentation | (ii) <i>Dianthus</i> |
| c. Basal placentation | (iii) <i>Argemone</i> |
| d. Free central placentation | (iv) Tomato |
- Choose the **correct** answer from the options given below.
- (1) a(iii), b(i), c(ii), d(iv) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(i), b(iii), c(ii), d(iv)

Space for Rough Work

108. In the equation $GPP - R = NPP$

R represents :

- (1) Respiration losses
- (2) Radiant energy
- (3) Retardation factor
- (4) Environmental factor

109. The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called

- (1) Wenn diagram (2) Pie diagram
- (3) A pyramid diagram (4) Punnett square

110. Which of the following is an **incorrect** statement?

- (1) Nuclear pores are the passages for proteins and RNA molecules and their movement takes place in both directions between nucleus and cytoplasm
- (2) Mature sieve tube elements of vascular plants possess a conspicuous nucleus
- (3) Microbodies are present in both plant and animal cells
- (4) The nuclear envelope forms a barrier between the materials present inside the nucleus and that of the cytoplasm

111. Match **List-I** with **List-II**.

List - I

- a. Lenticels
- b. Cork cambium
- c. Secondary cortex
- d. Cork

List - II

- (i) Phellogen
- (ii) Suberin deposition
- (iii) Exchange of gases
- (iv) Phelloderm

Choose the **correct** answer from the options given below

- (1) a(iv), b(ii), c(i), d(iii) (2) a(iv), b(i), c(iii), d(ii)
- (3) a(iii), c(i), c(iv), d(ii) (4) a(ii), b(iii), c(iv), d(i)

112. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as

- (1) Standing crop (2) Sere
- (3) Climax community (4) Standing state

113. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is

- (1) Cleistogamy (2) Xenogamy
- (3) Geitonogamy (4) Chasmogamy

114. Which of the following stages of meiosis involve division of centromere?

- (1) Telophase II (2) Metaphase I
- (3) Metaphase II (4) Anaphase II

115. Which of the following is the **correct** sequence of steps in a polymerase chain reaction?

- (1) Annealing → Denaturation → Extension
- (2) Denaturation → Annealing → Extension
- (3) Denaturation → Extension → Annealing
- (4) Extension → Denaturation → Annealing

116. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as

- (1) Safety testing (2) Biopiracy
- (3) Gene therapy (4) Molecular diagnosis

117. Match **List-I** with **List-II**.

List - I

- a. Cristae
- b. Thylakoid
- c. Centromere
- d. Cisternae

List - II

- (i) Primary constriction in chromosome
- (ii) Disc-shaped sacs in Golgi apparatus
- (iii) Infoldings in mitochondria
- (iv) Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- (1) a(ii), b(iii), c(iv), d(i) (2) a(iv), b(iii), c(ii), d(i)
- (3) a(i), b(iv), c(iii), d(ii) (4) a(iii), b(iv), c(i), d(ii)

Space for Rough Work

118. The factor that leads to founder's effect in a population is
 (1) Genetic drift (2) Natural selection
 (3) Genetic recombination (4) Mutation
119. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as
 (1) Acrocentric (2) Metacentric
 (3) Telocentric (4) Sub-metacentric
120. Gemmae are present in
 (1) Some liverworts (2) All mosses
 (3) Pteridophytes (4) Some gymnosperms
121. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out
 (1) Polysaccharides (2) RNA
 (3) DNA (4) Histones
122. Match **List-I** with **List-II**.
- | List-I | | List-II |
|--------------------|-------|--|
| a. Ethylene | (i) | Stimulates adventitious shoot formation |
| b. Absciscic acid | (ii) | Speed-up the malting process in brewing industry |
| c. GA ₃ | (iii) | Promotes female flowers in cucumbers |
| d. Cytokinins | (iv) | Stimulates the closure of stomata |
- Choose the **correct** answer from the option given below.
- (1) a(iii), b(i), c(iv), d(ii) (2) a(iii), b(iv), c(ii), d(i)
 (3) a(i), b(iii), c(iv), d(ii) (4) a(iv), b(i), c(ii), d(iii)
123. The site of perception of light in plants during photoperiodism is
 (1) Leaf (2) Shoot apex
 (3) Stem (4) Axillary bud
124. The plant hormone used to kill dicotyledonous weeds in a field is
 (1) IBA (2) IAA
 (3) NAA (4) 2, 4-D
125. Which of the following algae produce carrageen?
 (1) Blue-green algae (2) Green algae
 (3) Brown algae (4) Red algae
126. The diagram shows an important concept in the genetic implication of DNA. Fill in the blanks for a to d.
- (a) $\text{DNA} \xrightarrow{(b)} \text{mRNA} \xrightarrow{(c)} (d)$
- (1) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein
 (2) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
 (3) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Protein
 (4) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
127. Which of the following are not secondary metabolites in plants?
 (1) Rubber, gums (2) Morphine, codeine
 (3) Amino acids, glucose (4) Vinblastin, curcumin
128. In spite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival?
 (1) Predation (2) Resource partitioning
 (3) Competitive release (4) Mutualism
129. Which of the following is **not** an application of PCR?
 (1) Detection of gene mutation
 (2) Molecular diagnosis
 (3) Gene amplification
 (4) Purification of isolated protein

130. Match **List-I** with **List-II**.

List - I

- a. Cells with active cell division capacity
- b. Tissue having all cells similar in structure and function
- c. Tissue having different types of cells
- d. Dead cells with highly thickened walls and narrow lumen

List - II

- (i) Vascular tissues
- (ii) Meristematic tissue
- (iii) Sclereids
- (iv) Simple tissue

Select the **correct** answer from the options given below.

- (1) a(iii), b(ii), c(iv), d(i) (2) a(ii), b(iv), c(i), d(iii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(i), b(ii), c(iii), d(iv)

131. A typical angiosperm embryo sac at maturity is

- (1) 8-nucleate and 8-celled
- (2) 8-nucleate and 7-celled
- (3) 7-nucleate and 8-celled
- (4) 7-nucleate and 7-celled

132. Which of the following algae contains mannitol as reserve food material?

- (1) *Ulothrix* (2) *Ectocarpus*
(3) *Gracilaria* (4) *Volvox*

133. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as

- (1) Heterosporous (2) Homosorus
(3) Heterosorus (4) Homosporous

134. Mutations in plant cells can be induced by

- (1) Zeatin (2) Kinetin
(3) Infrared rays (4) Gamma rays

135. Amensalism can be represented as

- (1) Species A (+); Species B (0)
- (2) Species A (–); Species B (0)
- (3) Species A (+); Species B (+)
- (4) Species A (–); Species B (–)

SECTION-B

136. What is the role of RNA polymerase III in the process of transcription in eukaryotes?

- (1) Transcribes only snRNAs
- (2) Transcribes rRNAs (28S, 18S and 5.8S)
- (3) Transcribes tRNA, 5s rRNA and snRNA
- (4) Transcribes precursor of mRNA

137. Select the **correct** pair.

- (1) Loose parenchyma cells – Spongy rupturing the epidermis parenchyma and forming a lens shaped opening in bark
- (2) Large colorless empty cells in the epidermis of grass leaves --- Subsidiary cells
- (3) In dicot leaves, vascular bundles are surrounded by large thick-walled cells --- Conjunctive tissue
- (4) Cells of medullary rays that form part of cambial ring --- Interfascicular cambium

138. Plasmid pBR322 has *Pst* I restriction enzyme site within gene *amp^R* that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactosidase production and the recombinant plasmid is inserted in an *E.coli* strain

- (1) It will be able to produce a novel protein with dual ability
- (2) It will not be able to confer ampicillin resistance to the host cell
- (3) The transformed cells will have the ability to resist ampicillin as well as produce β -galactosidase
- (4) it will lead to lysis of host cell

139. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as

- (1) Polymorphic DNA (2) Satellite DNA
(3) Repetitive DNA (4) Single nucleotides

Space for Rough Work

140. In the exponential growth equation $N_t = N_0 e^{rt}$, e represents

- (1) Intrinsic rate of natural increase
- (2) Population density at time zero
- (3) Population density after time 't'
- (4) The base of natural logarithms

141. Which of the following statements is **incorrect**?

- (1) Cyclic photophosphorylation involves both PS I and PS II
- (2) Both ATP and NADPH + H^+ are synthesized during non-cyclic photophosphorylation
- (3) Stroma lamellae have PS I only and lack NADP reductase
- (4) Grana lamellae have both PS I and PS II

142. Which of the following statements is **correct**?

- (1) Some of the cyanobacteria can fix atmospheric nitrogen in specialized cells called bundle sheath cells
- (2) Fusion of two cells is called Karyogamy
- (3) Fusion of protoplasts between two motile or non-motile gametes is called plasmogamy
- (4) Organisms that depend on living plants and animals are called saprophytes

143. Match Column-I with Column-II.

Column-I	Column-II
a. Viroids	(i) Flagellated protozoan, causes sleeping sickness
b. <i>Albugo</i>	(ii) Mycelium is septate and branched
c. <i>Trypanosoma</i>	(iii) Free RNA, causes potato spindle tuber disease
d. <i>Claviceps</i>	(iv) Mycelium is aseptate and coenocytic

Choose the **correct** answer from options given below.

- (1) a(ii), b(i), c(iv), d(iii)
- (2) a(i), b(ii), c(iv), d(iii)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(iii), b(iv), c(i), d(ii)

144. Which of the following statements is **incorrect**?

- (1) Oxidation-reduction reactions produce proton gradient in respiration
- (2) During aerobic respiration, role of oxygen is limited to the terminal stage
- (3) In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and one $FADH_2$ gives rise to 3 ATP molecules
- (4) Complex V is ATP synthase complex

145. In some members of which of the following pairs of families, pollen grains retain their viability for months after release?

- (1) Rosaceae ; Leguminosae
- (2) Poaceae ; Rosaceae
- (3) Poaceae ; Leguminosae
- (4) Poaceae ; Solanaceae

146. Identify the **correct** statement.

- (1) Split gene arrangement is characteristic of all prokaryotes
- (2) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA
- (3) RNA polymerase associates transiently with Rho factor to terminate the process of transcription in bacteria
- (4) The coding strand in a transcription unit is copied to an mRNA

147. Match Column-I with Column-II.

Column - I	Column - II
a. $\% \frac{K_{(5)} C_1 + 2 + (2) A_{(2)} + \frac{G}{1}}{100}$	(i) Brassicaceae
b. $\frac{K_{(5)} C_{(5)} A_{(5)} S_{(2)}}{100}$	(ii) Liliaceae
c. $\frac{B \cdot \frac{P}{100} + 2 + (2) A_{(2)} + \frac{G}{100}}{100}$	(iii) Fabaceae
d. $\frac{K_{(2)} + 2 + C_2 A_{(2)} + \frac{S}{100}}{100}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

- (1) a(iv), b(ii), c(i), d(iii)
- (2) a(iii), b(iv), c(ii), d(i)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(ii), b(iii), c(iv), d(i)

148. Match List-I with List-II.

List - I	List - II
a. Protein	(i) C = C double bonds
b. Unsaturated fatty acid	(ii) Phosphodiester bonds
c. Nucleic acid	(iii) Glycosidic bonds
d. Polysaccharide	(iv) Peptide bonds

Choose the **correct** answer from the options given below.

- (1) a(iv), b(iii), c(i), d(ii) (2) a(iv), b(i), c(ii), d(iii)
 (3) a(i), b(iv), c(iii), d(ii) (4) a(ii), b(i), c(iv), d(iii)

149. Match List-I with List-II.

List - I	List - II
a. S phase	(i) Proteins are synthesized
b. G ₂ phase	(ii) Inactive phase
c. Quiescent stage	(iii) Interval between mitosis and initiation of DNA replication
d. G ₁ phase	(iv) DNA replication

Choose the **correct** answer from the options given below.

- (1) a(ii), b(iv), c(iii), d(i) (2) a(iii), b(ii), c(i), d(iv)
 (3) a(iv), b(ii), c(iii), d(i) (4) a(iv), b(i), c(ii), d(iii)

150. Nowadays, it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complementary DNA in a clone of cells, followed by its detection using autoradiography because

- (1) Mutated gene does not appear on photographic film as the probe has complementarity with it
 (2) Mutated gene partially appears on a photographic film
 (3) Mutated gene completely and clearly appears on a photographic film
 (4) Mutated gene does not appear on a photographic film as the probe has no complementarity with it

ZOOLOGY

SECTION-A

151. Erythropoietin hormone which stimulates RBC formation is produced by

- (1) Juxtaglomerular cells of the kidney
 (2) Alpha cells of pancreas
 (3) The cells of adenohypophysis
 (4) The cells of bone marrow

152. Which of the following characteristics is incorrect with respect to cockroach?

- (1) 10th abdominal segment in both sexes bears a pair of anal cerci.
 (2) A ring of gastric caeca is present at the junction of midgut and hindgut.
 (3) Hypopharynx lies within the cavity enclosed by the mouth parts.
 (4) In females, 7th-9th sterna contributes in the formation of genital pouch.

153. In a male frog, the Bidder's canal is present in the

- (1) Testes (2) Kidneys
 (3) Vasa efferentia (4) Cloaca

154. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature?

- (1) Pachytene (2) Leptotene
 (3) Zygotene (4) Diakinesis

155. The partial pressures (in mm Hg) of oxygen (O₂) and carbon dioxide (CO₂) at alveoli (the site of diffusion) are

- (1) pO₂ = 159 and pCO₂ = 0.3
 (2) pO₂ = 104 and pCO₂ = 40
 (3) pO₂ = 40 and pCO₂ = 45
 (4) pO₂ = 95 and pCO₂ = 40

Space for Rough Work

156. Match **List-I** with **List-II**.

List - I **List - II**

- | | |
|-----------------------|----------------------------|
| a. <i>Physalia</i> | (i) Pearl oyster |
| b. <i>Limulus</i> | (ii) Portuguese man of war |
| c. <i>Ancylostoma</i> | (iii) Living fossil |
| d. <i>Pinctada</i> | (iv) Hookworm |

Choose the **correct** answer from the options given below.

- (1) a(i), b(iv), c(iii), d(ii) (2) a(ii), b(iii), c(i), d(iv)
 (3) a(iv), b(i), c(iii), d(ii) (4) a(ii), b(iii), c(iv), d(i)

157. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first?

- (1) Polymerisation (2) Annealing
 (3) Extension (4) Denaturation

158. Match **List-I** with **List-II**.

List - I **List - II**

- | | |
|---------------------------------|-------------------|
| a. <i>Aspergillus niger</i> | (i) Acetic Acid |
| b. <i>Acetobacter aceti</i> | (ii) Lactic Acid |
| c. <i>Clostridium butylicum</i> | (iii) Citric Acid |
| d. <i>Lactobacillus</i> | (iv) Butyric Acid |

Choose the **correct** answer from the options given below.

- (1) a(iv), b(ii), c(i), d(iii) (2) a(iii), b(i), c(iv), d(ii)
 (3) a(i), b(ii), c(iii), d(iv) (4) a(ii), b(iii), c(i), d(iv)

159. Which of the following statements **wrongly** represents the nature of smooth muscles?

- (1) These muscles are present in the wall of blood vessels.
 (2) Fibres of these muscle have no striations.
 (3) They are involuntary muscles.
 (4) Communication among the cells is performed by intercalated discs.

160. Chronic auto immune disorder affecting the neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called

- (1) Gout (2) Arthritis
 (3) Muscular dystrophy (4) Myasthenia gravis

161. Persons with 'AB' blood group are called 'Universal recipients'. This is due to

- (1) Absence of antibodies anti-A and anti-B in plasma
 (2) Absence of antigens A and B on the surface of RBCs
 (3) Absence of antigens A and B in the plasma
 (4) Presence of antibodies, anti-A and anti-B on RBCs

162. The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase?

- (1) 32 (2) 8
 (3) 16 (4) 4

163. Receptors for sperm binding in mammals are present on

- (1) Zona pellucida
 (2) Corona radiata
 (3) Vitelline membrane
 (4) Perivitelline space

164. Which one of the following belongs to the family Muscidae?

- (1) House fly (2) Fire fly
 (3) Grasshopper (4) Cockroach

165. Select the **incorrect** pair.

- (1) Drug – Ricin
 (2) Alkaloid – Codeine
 (3) Toxin – Abrin
 (4) Lectin – Concanavalin A

166. Venereal diseases can spread through

- (a) Using sterile needles
 (b) Transfusion of blood from infected person
 (c) Infected mother to foetus
 (d) Kissing
 (e) Inheritance

Choose the **correct** answer from the options given below.

- (1) (a) and (c) only (2) (a), (b) and (c) only
 (3) (b), (c) and (d) only (4) (b) and (c) only

Space for Rough Work

167. The organelles that are included in the endomembrane system are
 (1) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes
 (2) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
 (3) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
 (4) Golgi complex, Mitochondria, Ribosomes and Lysosomes
168. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following techniques is not useful for early detection?
 (1) PCR
 (2) Serum analysis
 (3) Recombinant DNA technology
 (4) ELISA Technique
169. Read the statements given below w.r.t insulin.
 (a) C-peptide is not present in mature insulin.
 (b) The insulin produced by rDNA technology has C-peptide.
 (c) The pro-insulin has a C-peptide.
 (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.
 Choose the option with **correct** statements only.
 (1) (a) and (d) only (2) (b) and (d) only
 (3) (b) and (c) only (4) (a), (c) and (d) only
170. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is
 (1) Poly(A) tail sequences
 (2) Degenerate primer sequences
 (3) Okazaki sequences
 (4) Palindromic nucleotide sequences
171. Which is the "Only enzyme" that has "Capability" to catalyse the transcription of mRNA, tRNA, rRNA in prokaryotes?
 (1) DNase
 (2) DNA dependent DNA polymerase
 (3) DNA dependent RNA polymerase
 (4) DNA Ligase
172. Which one of the following organisms bears hollow and pneumatic long bones?
 (1) *Ornithorhynchus* (2) *Neophron*
 (3) *Hemidactylus* (4) *Macropus*
173. Read the following statements
 (a) Metagenesis is observed in helminths.
 (b) Echinoderms are triploblastic and coelomate animals.
 (c) Roundworms have organ-system level of body organization.
 (d) Comb plates present in ctenophores help in digestion.
 (e) Water vascular system is characteristic of echinoderms.
 Choose the **correct** answer from the options given below.
 (1) (b), (c) and (e) are correct
 (2) (c), (d) and (e) are correct
 (3) (a), (b) and (c) are correct
 (4) (a), (d) and (e) are correct
174. Match **List-I** with **List-II**.
- | List - I | List - II |
|-----------------|--|
| a. Vaults | (i) Entry of sperm through cervix is blocked |
| b. IUDs | (ii) Removal of a small part of vas deferens |
| c. Vasectomy | (iii) Phagocytosis of sperms within the uterus |
| d. Tubectomy | (iv) Removal of a small part of fallopian tube |
- Choose the **correct** answer from the options given below.
 (1) a(iii), b(i), c(iv), d(ii) (2) a(iv), b(ii), c(i), d(iii)
 (3) a(i), b(iii), c(ii), d(iv) (4) a(ii), b(iv), c(iii), d(i)
175. Choose the **incorrect** option w.r.t. frogs.
 (1) They contain a pair of nostrils above the mouth.
 (2) Their webbed digits facilitate swimming.
 (3) They have digestive glands like pancreas and liver.
 (4) Their intestine is devoid of villi and microvilli.

Space for Rough Work

176. If adenine makes 30% of the DNA molecule, what will be the percentage of thymine, guanine and cytosine in it?

- (1) T : 20 ; G : 25 ; C : 25
 (2) T : 20 ; G : 30 ; C : 20
 (3) T : 20 ; G : 20 ; C : 30
 (4) T : 30 ; G : 20 ; C : 20

177. Match **List-I** with **List-II**.

List - I**List - II**

- | | |
|-----------------|------------------|
| a. Metamerism | (i) Coelenterata |
| b. Canal system | (ii) Ctenophora |
| c. Comb plates | (iii) Annelida |
| d. Cnidoblasts | (iv) Porifera |

Choose the **correct** answer from the options given below.

- (1) a(iv), b(i), c(ii), d(iii) (2) a(iv), b(iii), c(i), d(ii)
 (3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(iv), c(ii), d(i)

178. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins?

- (1) Thrombokinase (2) Thrombin
 (3) Renin (4) Epinephrine

179. Which of the following is **not** true about tropical regions?

- (1) More solar energy is available contributing higher productivity
 (2) Promotes niche specialisation
 (3) Relatively more constant and predictable
 (4) Less rainfall and subjected to frequent glaciations in the past

180. A biocontrol agent that is very common in root ecosystems and used in the treatment of plant diseases is

- (1) Dragonflies
 (2) Ladybird
 (3) *Trichoderma*
 (4) *Bacillus thuringiensis*

181. Which one of the following is an example of a hormone releasing IUD?

- (1) Multiload 375 (2) CuT
 (3) LNG 20 (4) Cu 7

182. The centriole undergoes duplication during

- (1) G₂ phase (2) S-phase
 (3) Prophase (4) Metaphase

183. Which of the following RNAs is **not** required for the synthesis of protein?

- (1) siRNA (2) mRNA
 (3) tRNA (4) rRNA

184. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased?

- (1) 100% (2) 50%
 (3) 75% (4) 25%

185. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

- (1) Low pO₂, low pCO₂, more H⁺, higher temperature
 (2) High pO₂, low pCO₂, less H⁺, lower temperature
 (3) Low pO₂, high pCO₂, more H⁺, higher temperature
 (4) High pO₂, high pCO₂, less H⁺, higher temperature

SECTION-B

186. Match **List-I** with **List-II**.

List - I**List - II**

- | | |
|-----------------------------|----------------------------|
| a. Allen's Rule | (i) Kangaroo rat |
| b. Physiological adaptation | (ii) Desert lizard |
| c. Behavioural adaptation | (iii) Marine fish at depth |
| d. Biochemical adaptation | (iv) Polar sea |

Choose the **correct** answer from the options given below.

- (1) a(iv), b(iii), c(ii), d(i) (2) a(iv), b(ii), c(iii), d(i)
 (3) a(iv), b(i), c(iii), d(ii) (4) a(iv), b(i), c(ii), d(iii)

Space for Rough Work

187. Match **List-I** with **List-II**.

- | List - I | List - II |
|--------------------------------------|--|
| a. Adaptive radiation | (i) Selection of resistant varieties due to excessive use of herbicides and pesticides |
| b. Convergent evolution | (ii) Bones of forelimbs in man and whale |
| c. Divergent evolution | (iii) Wings of butterfly and bird |
| d. Evolution by anthropogenic action | (iv) Darwin's finches |

Choose the **correct** answer from the options given below.

- (1) a(i), b(iv), c(iii), d(ii) (2) a(iv), b(iii), c(ii), d(i)
 (3) a(iii), b(ii), c(i), d(iv) (4) a(ii), b(i), c(iv), d(iii)

188. Which of these is **not** an important component of initiation of parturition in humans?

- (1) Release of prolactin
 (2) Increase in estrogen and progesterone ratio
 (3) Synthesis of prostaglandins
 (4) Release of oxytocin

189. Which one of the following statements about Histones is **wrong**?

- (1) Histones carry positive charge in the side chain
 (2) Histones are organized to form a unit of 8 molecules
 (3) The pH of histones is slightly acidic
 (4) Histones are rich in amino acids - Lysine and Arginine

190. Which of the following events occur during complete muscular contraction?

- (a) 'H' zone disappears
 (b) 'A' band widens
 (c) 'I' band reduces in width
 (d) Myosine hydrolyzes ATP, releasing the ADP and Pi.
 (e) Z-lines attached to actins are pulled inwards.

Choose the **correct** answer from the options given below:

- (1) (b), (d), (e), (a) only (2) (a), (c), (d), (e) only
 (3) (a), (b), (c), (d) only (4) (b), (c), (d), (e) only

191. Match **List-I** with **List-II** w.r.t humans.

- | List - I | List - II |
|---------------------|---------------------------|
| a. Scapula | (i) Cartilaginous joints |
| b. Cranium | (ii) Flat bone |
| c. Sternum | (iii) Fibrous joints |
| d. Vertebral column | (iv) Triangular flat bone |

Choose the **correct** answer from the options given below.

- (1) a(iv), b(iii), c(ii), d(i) (2) a(i), b(iii), c(ii), d(iv)
 (3) a(ii), b(iii), c(iv), d(i) (4) a(iv), b(ii), c(iii), d(i)

192. **Assertion (A):** A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R): Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) (A) is false but (R) is true
 (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (4) (A) is true but (R) is false

193. The adenosine deaminase deficiency results into

- (1) Addison's disease
 (2) Dysfunction of immune system
 (3) Parkinson's disease
 (4) Digestive disorder

194. Select the **incorrect** option w.r.t. brain of frogs.

- (1) Unpaired diencephalon is present in forebrain.
 (2) Midbrain contains olfactory lobes.
 (3) Hindbrain consists of cerebellum.
 (4) Medulla oblongata passes out through the foramen magnum.

Space for Rough Work

195. Match **List-I** with **List-II**.

List - I

List - II

- a. Filariasis (i) *Haemophilus influenzae*
b. Amoebiasis (ii) *Trichophyton*
c. Pneumonia (iii) *Wuchereria bancrofti*
d. Ringworm (iv) *Entamoeba histolytica*

Choose the **correct** answer from the options given below.

- (1) a(ii), b(iii), c(i), d(iv) (2) a(iv), b(i), c(iii), d(ii)
(3) a(iii), b(iv), c(i), d(ii) (4) a(i), b(ii), c(iv), d(iii)

196. Given below are the statements with reference to 'lipids'.

- (a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including the carboxyl carbon.
(e) Arachidonic acid has 16 carbon atoms.

Choose the option with **correct** statements only.

- (1) (b) and (e) only (2) (a) and (b) only
(3) (c) and (d) only (4) (b) and (c) only

197. Following are the statements about prostomium of earthworm.

- (a) It serves as a covering for mouth.
(b) It helps to open cracks in the soil into which it can crawl.
(c) It is one of the sensory structures.
(d) It is the first body segment.

Choose the **correct** answer from the options given below.

- (1) Only (b) and (c) are correct
(2) (a), (b) and (c) are correct
(3) (a), (b) and (d) are correct
(4) (a), (b), (c) and (d) are correct

198. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy?

- (1) Uterus (2) Fallopian tube
(3) Ovary (4) Foetus

199. **Statement I:** The codon 'AUG' codes for methionine and phenylalanine.

Statement II: 'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Statement I is incorrect but Statement II is correct
(2) Both Statement I and Statement II are correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is correct but Statement II is incorrect

200. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules respectively.

- (1) Adhering junctions and gap junctions
(2) Gap junctions and adhering junctions
(3) Tight junctions and gap junctions
(4) Adhering junctions and tight junctions



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